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ORIGINAL RESEARCH

Measuring the Long and Healthy Lives of a Nation: the Health, Aging and Gender Nexus

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Abstract
That a nation's populace lives Long and Healthy Lives (LHL) is essential for its economic functioning and wellbeing. The United Nations Development Program (UNDP)'s life expectancy at birth focused LHL Index, contrary to current evidence on the health-age relationship and the health-longevity gender paradox, assumes constant health outcomes over life stage and gender divides and usually finds intertemporal improvements with superior results for women in most nations. Here, techniques that respond to ordinal data scaling critiques are developed and employed to provide indices that incorporate life stage health information. The indices are introduced in 21st Century applications to China, UK, and USA. In accord with current evidence, much variation in health outcomes over life stage and gender divides is revealed and the usual UNDP findings are reversed with intertemporal deterioration in LHL and inferior results for women, a composition effect largely the consequence of an aging population.

Keywords Health inequality · Long and healthy lives measurement · Health-longevity paradox

1 Introduction

A fundamental tenet of the Functionings and Capabilities Approach (Nussbaum & Sen, 1993; Sen, 1999) to wellbeing, and an integral part of a nation's economic growth and development, is that its constituents live long and healthy lives. The approach was foundational in developing the United Nations Development Programme's Long and Healthy Life index (UNDP, 2019), a cornerstone of its Human Development Index (HDI). On the twin

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implicit presumptions that perceived health status is constant across the life cycle, and that a longer life implies a healthier lived life (which renders longevity a good proxy for lifetime health outcomes), the "Long and Healthy Life" measure, is based solely upon Life Expectancy at birth (UNDP, (2019). However, such presumptions can be misleading since it has long been understood that health outcomes deteriorate with age (Deaton and Paxson, 1998; Kerkhofs & Lindeboom, 1997; Miller and Canning 2019). Furthermore, recent work (Case and Paxson, 2005; Nusselder et al., 2010; Oksuzyan et al., 2009; Van Oyen et al., 2013) has highlighted a health-longevity gender paradox, that throughout their lives women typically experience inferior health outcomes at comparable life stages yet tend to live longer than men. Basically, in spite of females having more bad health markers (sequelae) than males which engender worse health outcomes at every life stage, their average age at death (life expectancy) is significantly higher than males. The distinction is of paramount importance in the context of an aging society as is the case in three societies considered in this study (China, UK and USA). In the decade covered by this study, fertility rates in the three countries have been persistently below 2.1, the number deemed necessary for a non-aging population, with respective averages of 1.66, 1.80 and 1.85. Low fertility rates together with the gender paradox cause the proportion of older, less healthy individuals to increase and the proportion of younger, more healthy individuals to diminish differentially across the gender divide. The overall result is an increase in the dependency ratio with a concomitant increase in the society's health care burden, both in terms of the fiduciary concerns (younger cohorts are the primary tax base and older cohorts are the primary transfer recipients) and real factor concerns (younger cohorts are the primary suppliers of care and older cohorts are the primary demanders of care).

These concerns argue for an index which modulates the future life length a group can reasonably expect to live by the health status they currently enjoy at their given life stage. In recognition of this, based upon a complex and intensive Monte Carlo simulation analysis of over 1000 sequelae of disease and injury in 187 countries, Salomon et al. (2012), produced the Healthy Life Expectancy (HALE) index, a summary measure of population health combining mortality information from a period life table with cross-sectional information on the prevalence of morbidity or actual health conditions in the population of interest[1]. HALE is best interpreted as the number of years a person can expect to live in good health, but it doesn't account for the changing demographic of proportionately increasing elderly populations and proportionately diminishing younger populations brought about by low fertility rates. However, the Functionings and Capabilities Approach would argue that, at each stage of the life cycle, beyond the actuality of their physical and mental health, it is an individual's perception of their health status that impedes or enhances their ability to be and do and live the life they desire (see for example Machon et al., 2016; Tovel et al., 2019; Miller and Bairoliya, 2021). Given that it is an individual's perception of their health status that matters, a nation's long and healthy life index based upon individual perceptions of

[1] The Global Burden of Disease Study (GBD) 2010 upon which HALE is based, provides estimates of the prevalence of 1160 sequelae (a health condition which is the consequence of previous disease and injury) based on systematic analyses of all available data for 187 countries in 1990 and 2010, along with new estimates of disability weights representing the magnitude of health losses associated with each sequelae.

health status at given stages of the lifecycle which are comparable across nations that would be appropriate[2].

Self-reported health status is invariably ordinal in nature and dependent upon the nature of the questionnaire, which makes comparability across jurisdictions problematic. Analysis has frequently been facilitated by attributing cardinal scale to ordered categories (Likert, 1932; Rankin, 1957; Cantril, 1965) which led Bond and Lang (2019) and Schroder and Yitzhaki (2017) to reveal a "Sad Truth about Happiness Scales", that there is inherent ambiguity and equivocation in the measurement process engendered by scale dependency issues. Basically, equally valid alternative scales applied to a survey can yield substantively different estimates of levels and differences. However, by entertaining the concept of Probabilistic Distance[3], unequivocal and unambiguous scale independent cardinal measures of group health levels measured in the context of an aggregation of individual chances of better health outcomes are possible which obviate arbitrary attribution of cardinal measure to ordered categories. Indeed, such measures are closely related to the cardinal paradigm mean which can be written as the cumulation of chances of higher-level outcomes over the range of the variable. The measures are cast in terms of the probability of an individual being at a nominated or lower health level (given by the Cumulative Distribution Function of the reference population[4]) and its complement, the chance of achieving better than the nominated health level (given by the Survival Function). The cumulated chances of better outcomes can then be used as a measure of the level of health in the population, analogous to the mean of a cardinally measured variable. This measure can then be used to integrate measures of health status at a given age and gender group with measures of expected future life span for those groups to produce workable and simple "perception of health" based Long and Healthy Life indices.

To be clear, in the ordinal framework cardinality is imparted solely by the cumulation of cardinally measured probabilities to give a "chance of a healthier life" based index which is the sum over all reported health levels of the chance of a better health outcome at each level, whereas in the cardinal paradigm it is the cumulation of those probabilities scaled by the metric of the cardinally measured variable. In the ordinal case, the measure is bounded by the number of categories, comparison across jurisdictions with differing numbers of categories is facilitated by dividing by the number of categories minus one to give an average chance of a better outcome measure. Moreover, since the probabilistic distance measure is a metric common to any ordered categorical variable, comparison of distinctly different variables with different numbers of categories across more than one dimension is readily facilitated[5].

[2] Comparison of HALE, which reflects the actualities of an individual's health status, with self-reported perceptions of health status, which may well reflect prevalent social norms and customs, would be an interesting direction for future research, but it is beyond the scope of the current endeavor.

[3] As the basis of the Pearson (1900) family of Goodness of Fit Tests and Gini (1916) Transvariation measures which assess how far apart probability distributions are, Probabilistic Distance is an old idea, Mendelson (1987) revived and adapted it to measure quantile preserving spreads. It has seen some use in ordinal outcome contexts in the Health Inequality and Bi-Polarization literatures (Allison & Foster, 2004; Kobus, 2015; Kobus & Kurek, 2019; Anderson et al., 2021).

[4] The value of the Cumulative Distribution Function at an individual's health outcome level would provide an unambiguous – scale free indicator of its health level useful in regression analyses, but that possibility is not pursued here.

[5] Most importantly, these techniques are readily applicable in unambiguously and unequivocally quantifying levels of (and inferences about) other ordered categorical variables such as self-reported happiness, experi-

Aging populations with declining fertility or replication rates are common characteristics of many nations and a distinctive feature of the Chinese, UK and USA economies examined in this paper. Here, a new Long and Healthy Life index is developed and implemented for 21 st Century data in those countries. Initially, the study was based on Chinese data for 2008, (the year before substantial health care reforms were implemented) and for a selection of years thereafter up until 2017. Subsequently, for comparison purposes, analyses of UK and USA data over comparable periods were performed. Results across all three nations reveal that, consistent with the health-longevity gender paradox, females report less healthy outcomes than males but live longer lives. Health outcomes deteriorate significantly with age for both genders which, given ageing populations, places an increasing burden on health care provision. However, health levels have been maintained within gender and age-based groups. Finally, while the HDI health component indicates improving long and healthy lives with better outcomes for females than males, the Long and Healthy Life index developed here indicates the opposite with temporally deteriorating values that record inferior outcomes for females as opposed to males, the consequence of the deterioration of health with age, the health-longevity gender paradox and an aging population.

In the following, Sect. 2 outlines the methodology employed in analyzing and making inferences about ordered categorical data and proposes a new Long and Healthy Life Index. Section 3 reports the results of the analysis over a decade in 21 st Century China and, for comparison purposes, over similar periods for the United Kingdom and the United States and Sect. 4 draws conclusions.

2 Methodology

Since the health levels and unexpired life outcomes are ordered categorically, the attendant ambiguity and equivocation difficulties associated with measuring levels and differences in such ordinal data environments (Bond & Lang, 2019) will be surmounted by employing the construct of probabilistic distance (Mendelson, 1987). To understand the approach and highlight its similarities with those employed in cardinally measurable environment, the corresponding analysis of a continuously measured variable will be briefly outlined and the ordinal measures then developed by analogy.

2.1 The Cardinal Paradigm

To develop cardinal measures of levels in Ordered Categorical Paradigms, extensive use will be made of the Survival Function and its cumulants. In order to understand that relationship, the connection between the mean and the Survival Function in the continuous variable world is first outlined[6]. Suppose the quality of interest is a continuous variable x with $0 < x < Y < \infty$. Denote its group t population Probability Density Function (PDF) $f_t(x)$, its corresponding Cumulative Distribution Function (CDF) $P_t(X < x) = F_t(x) = \int_0^x f_t(z)\,dz$ and its Survival Function (SF) $P_t(X \geq x) = S_t(x) = 1 - F_t(x)$. Given an expectations opera-

ence, and embodied human capital etc.

[6] While the long and healthy life analysis is bivariate in nature, for expository reasons the discussion of the distributional aspects of the analysis is pursued in terms of a univariate problem on the understanding that the connections in a multidimensional world are similar (Atkinson & Bourguignon, (1982).

tor $E_{f_t(x)}$, its mean will be $\mu_t = E_{f_t(x)}(x) = \int_0^Y x f_t(x)\,dx$ which, by letting $u(x) = x$ and $v'(x) = f_t(x)$ and using the integration by parts rule[7] can be shown to be the integral of $S_t(x)$ since:

$$\mu_t = \int_0^Y x f_t(x)\,dx = [xF_t(x)]_0^Y - \int_0^Y F_t(x)\,dx = \int_0^Y (1 - F_t(x))\,dx = \int_0^Y S_t(x)\,dx$$

Since $S_t(x)$ is the probability of a randomly selected individual in population t having a higher level than x, the mean can be interpreted as the cumulation of those "better than x" chances over the range of x.

2.2 Properties of the Underlying long and Healthy Life Wellbeing Measure

An essential component of the Functionings and Capabilities Approach to Wellbeing and Human Development is that a nation's members live long, healthy enabling lives (Nussbaum & Sen, 1993; Sen, 1999). Wellbeing in this regard will be a function of the expected remaining life length (L) and the health level they can expect to enjoy during that time (H). Assuming all agents have common preferences, the Long and Healthy Life Utility Function $U(L, H)$ will be monotonic increasing and non-separable in L and H characterizing the mutual dependency of health level and future life length, thus:

$$\frac{\partial U(L,H)}{\partial L} > 0; \frac{\partial U(L,H)}{\partial H} > 0 \,\&\, \frac{\partial^2 U(L,H)}{\partial L \partial H} \neq 0 \tag{1}$$

In this format, the young are the most well off in terms of life length left to live and the old are the least well off, so the age distribution will be reversed with a longevity axis in reverse order of age, in effect $L = \tau - age$ where τ is the maximum possible age in the nation with its axis running in descending order from τ to 0. Similarly, the maximum level of health that is possible in a society is denoted θ with a minimum possible level of 0. In this context, denoting the j'th period joint PDF: $f_j(L, H)$, CDF: $F_j(L, H)$ and SF: $S_j(L, H)\ for\ j = t, t+1$, if health and life length were continuously measured and a society's Long and Healthy Life status in time periods t and $t+1$ compared, interest would focus on the change in the expected values of $U(L, H)$. In particular, progress in the societies long and healthy life would be judged by whether:

$$E_{f_{t+1}(L,H)}(U(L,H)) - E_{f_t(L,H)}(U(L,H)) = \int_\tau^0 \int_0^\theta U(L,H)(f_{t+1}(L,H) - f_t(L,H))\,dLdH > 0.$$

Following Atkinson and Bourguignon (1982) the difference will be unambiguously positive when $F_{t+1}(L, H) \leq F_t(L, H)\ \forall\ L\ and\ H\ with < at\ least\ somewhere$ or alternatively when:

[7] $\int_0^Y u(x)\,v'(x)\,dx = [u(x)\,v(x)]_0^Y - \int_0^Y u'(x)\,v(x)\,dx.$

$$S_{t+1}(L,H) \geq S_t(L,H) \ \forall \ L \ and \ H \ with > at \ least \ somewhere. \tag{2}$$

Any index of the extent to which a nation lives long and healthy lives would need to reflect these properties so that, holding all else constant, a marginal increase in health at some level would increase the index and similarly a marginal increase in expected life length at some point, again holding all else constant, would increase the index.

In univariate versions of [2] above, $S_t(x)$ is employed in stochastic dominance comparisons to unambiguously order the distributions of different groups in accord with an underlying General Value of x function, $U(x)$.

$$\begin{gathered} For \ any \ U(x), \ with \ U'(x) > 0, \ E_{f_{t+1}(x)}(U(x)) \geq E_{f_t(x)}(U(x)) \\ whenever \ S_{t+1}(x) \geq S_t(x) \forall x \ with \ S_{t+1}(x) > S_t(x) \ for \ some \ x. \end{gathered} \tag{3}$$

Note this condition is stronger than simply requiring $\mu_{t+1} > \mu_t$ which only requires $\int_0^Y (S_{t+1}(x) - S_t(x))dx > 0$ with negative differences being outweighed by positive differences. When $\mu_{t+1} > \mu_t$, satisfaction of [3] can be examined by contemplating an **Ambiguity Index (AMB)** measuring the extent to which the difference between group $t+1$ and t are ambiguous:

$$AMB = 1 - \frac{\int_0^Y (S_{t+1}(x) - S_t(x))dx}{\int_0^Y |S_{t+1}(x) - S_t(x)|dx}$$

When $AMB = 0$ group t+1 is unambiguously better off than group t, with at least as good a chance of a higher x and higher $U(x)$ at every level and a better chance at some levels[8].

To be clear, when $\mu_{t+1} > \mu_t$ in a statistically significant sense, it will be the case that $\int_0^Y (S_{t+1}(x) - S_t(x))dx > 0$ but the difference, though statistically significant, is only unambiguous and unequivocal when $\int_0^Y (S_{t+1}(x) - S_t(x))dx = \int_0^Y |S_{t+1}(x) - S_t(x)|dx$, i.e. when at every level x the chances of a higher level x are at least as great under $f_{t+1}(x)$ as they are under $f_t(x)$. This latter condition adds an additional layer to the nature of the difference in that a non-negative difference is experienced at every level of x, an implicit inclusivity condition which avers that group $t+1$ is at least as well of as group t at every level x.

When health and longevity are cardinally measured in discrete intervals, the analysis is similar with integral operators being replaced by summation operators, but the analysis runs into difficulties when one or more of the variates is ordered categorical, without cardinal measure.

[8] In effect, the cumulative density of $t+1$ will be everywhere below or equal to that of t so that the probability of being at a given level or lower will always be the same as or less in $t+1$ than it is in t.

2.3 The Ordered Categorical Paradigm

When working with self-reported health outcomes, the variable X does not have cardinal measure, instead it is reported in a sequence of ordered outcome categories. While, in order to facilitate analysis, there has long been a practice of attaching cardinal values to these ordinal categories (Cantril, 1965; Rankin, 1957; Likert, 1932), its equivocal and arbitrary nature has recently been the subject of criticism (Bond & Lang, 2019). However, it is possible to work with cumulated chances of better outcomes in an ordinal framework which yield unequivocal and unambiguous comparisons (see for example Anderson et al., 2020, 2022).

Suppose M ordered categories indexed $m = 1, .., M$, where the index reflects the ordering of the categories from least to most preferred, then the group t PDF is discrete ordered categorical, assigning probability $f_{m,t} \geq 0$ to category m such that $\sum_{m=1}^{M} f_{m,t} = 1$. Analogous to the continuous paradigm, $F_{m,t}$ is the CDF where $F_{m,t} = \sum_{j=1}^{m} f_{j,t}$ which is the chance of a randomly selected agent in the population being at an outcome level lower than or equal to m and the Survival Function $S_{m,t} = 1 - F_{m,t}$, $m = 1, .., M$. is the chance of a "better" outcome level than m for that agent. In matrix form let the Probability Density be $\underline{f}_t$, Cumulative Density be $\underline{f}_t$, and Survival Function vectors of group t respectively be $\underline{S}_t$ then:

$$\underline{f}_t = \begin{bmatrix} f_{1,t} \\ f_{2,t} \\ . \\ . \\ f_{M,t} \end{bmatrix} ; \underline{F}_t = \begin{bmatrix} F_{1,t} \\ F_{2,t} \\ . \\ . \\ F_{M,t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & . & 0 \\ 1 & 1 & 0 & . & 0 \\ 1 & 1 & 1 & . & 0 \\ . & . & . & . & 0 \\ 1 & 1 & 1 & . & 1 \end{bmatrix} \begin{bmatrix} f_{1,t} \\ f_{2,t} \\ . \\ . \\ f_{M,t} \end{bmatrix} \; and$$

$$\underline{S}_t = \begin{bmatrix} S_{1,t} \\ S_{2,t} \\ . \\ . \\ S_{M,t} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & . & 1 \\ 0 & 0 & 1 & . & 1 \\ 0 & 0 & 0 & . & 1 \\ . & . & . & . & 1 \\ 0 & 0 & 0 & . & 0 \end{bmatrix} \begin{bmatrix} f_{1,t} \\ f_{2,t} \\ . \\ . \\ f_{M,t} \end{bmatrix}$$

Even though there is no scaling in this environment, given an M dimensioned unit vector $\underline{i}_M$, the vector of better chances can still be cumulated to provide a cardinal measure of the extent of better chances in the population analogous to the "cumulated better chances" interpretation of the mean in the cardinal paradigm. This cumulation denoted $\mu_{OC,t}$ will be referred to as the **Ordered Categorical Mean (OCM)** where[9]:

$$\mu_{OC,t} = \underline{i}_M{}' \, \underline{S}_t \tag{4}$$

In the univariate ordered categorical environment, OCM will provide a measure and index of level (of health or longevity, for example) however in this environment the usual analytical calculus cannot be implemented i.e. though $\mu_{OC,t}$ is continuously measured in probability, the construct $\frac{d\mu_{OC,t}}{dx}$ cannot be contemplated for checking the monotonic increasing in x property demanded of an index because x does not have continuous cardinal value. However, in its place, the effect on $\mu_{OC,t}$ of an incremental upward transfer of probability mass $\Delta p > 0$ from any given category to the next highest category can be contemplated. In this context where j is a given category, dx_j implies $p_j - \Delta p$ and $p_{j+1} + \Delta p$. Letting

[9] Following Rao (2009) inference procedures employed in this context are based upon the asymptotic normality of [4] and are described in the appendix.

$df_{j,t}$ be an $M \times 1$ vector with $-\Delta p$ in its j'th element and $+\Delta p$ in its $j+1$'th element with zeros elsewhere, for any $j = 1, .., M-1$, $\frac{d\mu_{OC,t}}{dx_j}$ may be written as:

$$\frac{d\mu_{OC,t}}{dx_j} = i'_{-M} \left[\begin{bmatrix} 0 & 1 & 1 & . & 1 \\ 0 & 0 & 1 & . & 1 \\ 0 & 0 & 0 & . & 1 \\ . & . & . & . & 1 \\ 0 & 0 & 0 & . & 0 \end{bmatrix} \left[\begin{bmatrix} f_{1,t} \\ f_{2,t} \\ . \\ . \\ f_{M,t} \end{bmatrix} + df_{j,t} \right] - \begin{bmatrix} 0 & 1 & 1 & . & 1 \\ 0 & 0 & 1 & . & 1 \\ 0 & 0 & 0 & . & 1 \\ . & . & . & . & 1 \\ 0 & 0 & 0 & . & 0 \end{bmatrix} \begin{bmatrix} f_{1,t} \\ f_{2,t} \\ . \\ . \\ f_{M,t} \end{bmatrix} \right]$$

Since the inner product of rows from 1 to j of the pre-multiplying matrix with the $df_{j,t}$ vector will result in $\Delta p - \Delta p$ and net out the transfer and the $j+1$ row with the $df_{j,t}$ vector will result in Δp while the $j+i$ ($i > 1$) rows inner products with $df_{j,t}$will be unaffected by Δp, $\frac{d\mu_{OC,t}}{dx} = \Delta p > 0$, confirming the monotonic increasing property of the index with respect to the improving chances of better outcomes.

It is worth emphasizing that $\mu_{OC,t}$ is dependent upon M, the number of categories (its maximum possible value is $M-1$ which occurs when $f_{M,t} = 1$) so that comparison of ordered categorical variates with different numbers of categories is problematic. This can be rectified by dividing $\mu_{OC,t}$ by $M-1$ which gives a value between zero and one for a construct much like a relative mean value which would be comparable across variates with different numbers of categories. Since in the nation analyses in the application, the number of categories does not change over time, the mean will not be so adjusted in that case.

It should be noted that in an ordinal context the ambiguity index records the extent to which group t is enjoying at least as good a chance of better outcomes in every category as group t+1 this ensuring that the improvement is universal and inclusive.

2.4 A long and Healthy Life Index: A Proposal

The health level enjoyed by a society will be a weighted sum of the health levels enjoyed by the mutually exclusive and exhaustive subgroups of its populace. Suppose the subgroups were age-based categories and the self-reported health of an individual was a monotonic increasing function of the level that he/she could expect to enjoy for the rest of their life, then in a Long and Healthy Life index, younger groups health responses should count more than older groups since they will be associated with a longer expected future life span, i.e. the weight attached to a particular groups health outcome should reflect the relative length of life they have left to live.

Given J possible health outcomes arranged in increasing order of health quality and indexed $j = 1, .., J$ and K age groups indexed $k = 1, .., K$ ordered in terms of decreasing expected future life length (so that $k = 1$ is the youngest group with the greatest future life expectancy and $k = K$ is the oldest group with the least) then the bivariate survival function $S_{j,k}$ $j = 1, .., J$ $k = 1, .., K$ would provide an obvious Long and Healthy Life index $LHL = \sum_{j=1}^{J} \sum_{k=1}^{K} S_{j,k}$ which gives equal weight to each age group. However, the effects of longevity and health would be convoluted, and their separate effects obfuscated in such an index. Suppose age groups have respective population shares a_k, and let $v_k = \sum_{l=k}^{K} a_l$, $k = 1, .., K$. For some intuition, $v_1 = \sum_{l=1}^{K} a_l = 1$, representing the fact that the whole of the population in the youngest category has the potential for living the whole of its lifespan, $v_2 = \sum_{l=2}^{K} a_l$ would be the probabilistic proportion of the life span that the population in Age Group 2 can expect to live and so on until $v_K = \sum_{l=K}^{K} a_l = a_K$

would be the probabilistic proportion of the life span that the population in the last stage of life can expect to live. As the elements v correspond to Survival Functions, their sum or weighted sum has the property of being monotonic increasing in incremental upward transfers of probability mass. An ordered categorical probabilistic **Average Expected Future Life Length Index (AVEFLLI)** for the population with age group categories $k = 1, .., K$ can then be written as.

$$AVEFLLI = \sum_{k=1}^{K} a_k v_k$$

Then contemplate an expected future life length weighting function[10] w_k, $k = 1, .., K$ where:

$$0 < w_k = {v_k} \Big/ {\sum_{j=1}^{K} v_j} < 1 \; with \sum_{k=1}^{K} w_k = 1$$

Suppose the Health Index for age group k is H_k then $\boldsymbol{LHL}$, **a Long and Healthy Life index** that accounts for and separately identifies the effects of both longevity and differing health outcomes over the life cycle, would be:

$$LHL = \sum_{k=1}^{K} w_k H_k$$

Note that an improvement in longevity involves a transfer of mass from an older group to a younger group increasing w_k for low k and reducing w_k for high K so that when younger groups are healthier LHL will increase. An improvement in health for some group k will involve a transfer of mass from a low-level health outcome to a higher-level outcome which will increase its H_k value and, all else being held constant, it will increase the overall level of LHL.

One advantage of this measure is versatility. It can be applied to measure health outcomes with varying numbers of categories while remaining comparable. Dividing the index by $K - 1$ would standardize the index to the unit interval and makes groups with different categorical arrangements readily comparable. This transformation turns the index into "an average probability of a better outcome statistic". Furthermore, inference is straightforward as detailed in the appendix since it is in effect a normal random variable multiplied by a known constant.

Diagram 1 layout the construction process of the indices employed in the analysis. H_k represents the average health level of group k in the society. **AVEFLLI** denotes the society's average expected future life length. **LHL** reports the Long and Healthy Life index of the society which weights the health level of each subgroup by the remaining life length they have left to live. Finally, **AMB** is a measure of the extent to which outcomes have improved unequivocally over time.

[10] Note that the weighting function w can also be shown to be increasing in incremental upward transfers of probability mass.

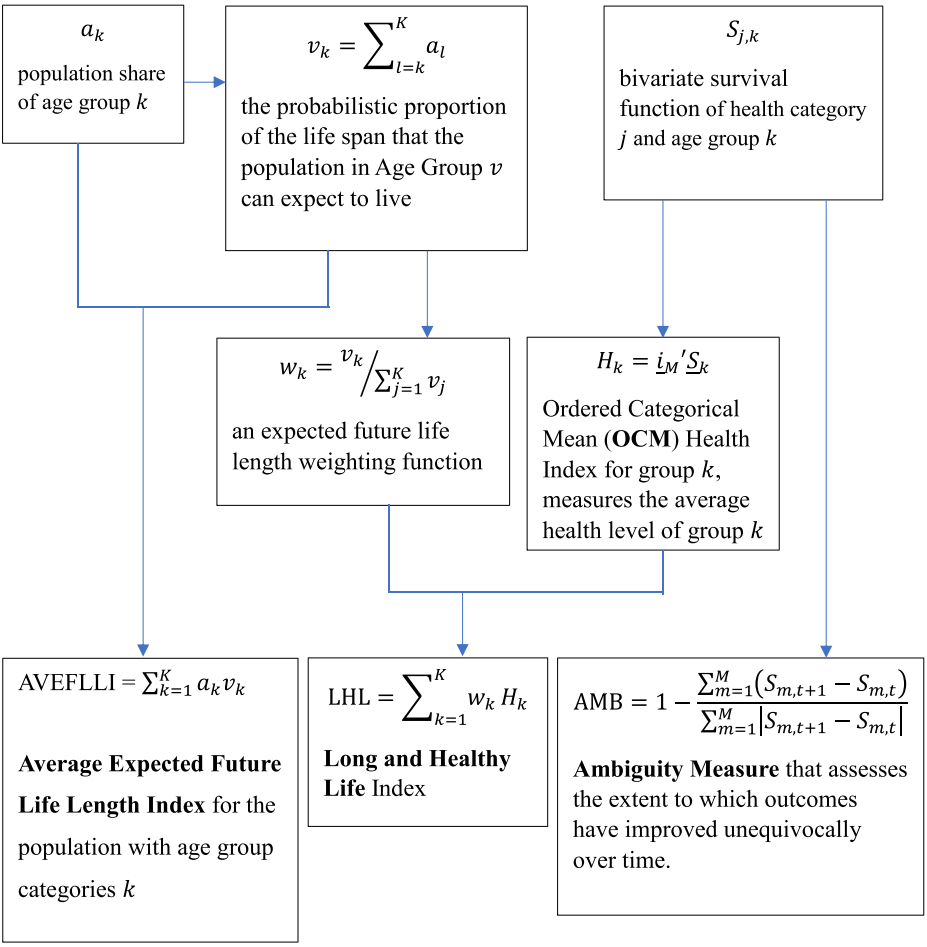

Diagram 1 Flowchart of Index Construction Process

3 Applications

In view of China's considerable public health care investments this century, this section examines its health and longevity status in detail. To exemplify the ability to compare groups with different category definitions, similar exercises are performed over a similar period (2010–2018) for the United States and the United Kingdom followed by a comparison of the 3 nations' outcomes is performed.

3.1 The Case of China

In 2009, after a sustained period of economic growth, China embarked upon major health-care reforms with the intent of providing all citizens with equal access to basic health care with reasonable quality and financial risk protection, especially for people of a lower socioeconomic status (Tao et al., 2020; Yip et al., 2019). The government not only expanded the scope and service package

of basic insurance coverage, but also increased the reimbursement rate. In 2016, the former urban and rural residents' health insurance systems UREBI (Urban Resident Basic Medical Insurance) and NRCMS (New Rural Cooperative Medical Scheme) were merged into the Urban and Rural Resident Medical Insurance, aiming to eliminate the rural and urban health gap (Tao et al., 2020). From 2008 to 2017, the government health-care expenditure quadrupled from 359 billion RMB to 1.52 trillion RMB, reaching 7.48% of the government general expenditure (Yip et al., 2019). Currently, though 70% was the target, about half the health care costs are covered publicly, leaving considerable room for the dependence of health outcomes on income status. This can be seen as a move from what was in essence a privately provided health care system, such as is the case in the United States, toward a publicly funded system such as is the case in the United Kingdom.

Self-reported health data is drawn from the 2008–2017 Chinese General Social Survey (Bian & Li, 2012), covering 26 Chinese provinces with complete and consistent data. Respondents were asked to report their current health status using the following five categories: 1 Very Unhealthy, 2 Relatively Unhealthy, 3 Normal, 4 Relatively Healthy, or 5 Very Healthy. The individual health outcomes can be summarized in six gender-based age cohorts so that cumulative densities for gender specific age cohorts could be readily calculated together with corresponding survival functions.

Table 1 and the following diagrams report the age group and gender specific Ordered Category Mean (OCM) health index together with the Average Expected Future Life Length Index (AVEFLLI) and the corresponding Long and Healthy Life index (LHL) for males, females and combined groups from 2008 to 2017. AVEFLLI is consistently lower for males than it is for females, a reflection of the fact that females live longer than males but note that it is deteriorating over time for both genders. This reflects the fact that China's low fertility rate is reducing the proportion of younger cohorts with a long life left to live and increasing the share of older cohorts with a shorter length of life left to live.

The gender specific AVEFLLI and LHL indices reported in Table 1 are presented graphically in Diagram 2 and 3, respectively. In Diagram 2, AVEFLLI trends highlight the decline in China's stock of future life length, a result of the proportionate decline of young who have a long expected future life length and an increase in the old with a short expected future life length ahead of them. As shown in Diagram 3, LHL is consistently higher for males than it is for females throughout the observation period. Nonetheless, contrary to what is observed with a purely Life Expectancy based index or the HALE index for China[11], the LHL indicates a significant temporal deterioration in the Long and Healthy Lives lived by males (−0.1272 s.e. 0.0185), females (−0.1022 s.e. 0.0180) and the overall population (−0.1146 s.e. 0.0124). This is to be expected when it is noted that a society's population is ageing and older groups have inferior health outcomes since it is quite simply a composition effect, groups with inferior health outcomes are proportionately increasing in the population. Notice such decline was interrupted by the impact of China's huge increased investment in health care provision post 2010.

Table 1 and diagram 4 highlight gender differences in OCM health outcomes over the lifecycle. Notice the systematic decline in health over the life cycle where each age group except the oldest, males have better outcomes than females. While there is some variation in within gender based age group outcomes over the period, it is never substantive, which is a testament to China's efforts in maintaining the health levels of its citizenry.

[11] The World Bank reports respective 2008 and 2017 life expectancies in China of 72 and 74 for males, 78 and 80 for females and 75 and 77 overall, and Chen et al. (2022) report increases in HALE for China over the 1990–2019 period.

Table 1 China Average Expected Life Length and Long and Healthy Life Indices, 2008–2017

Year & Sample size	Male & Female Share	Age Cohort OCM Health Indices, (Standard Errors in italics.)						AVEFLLI and LHL Indices			
		Age Group									
		<25	25–34	35–44	45–54	55–64	>64		Males	Females	Overall
2008	M%=48.21%	3.2461	2.9694	2.7262	2.4707	2.1911	1.9545	AVEFLLI	4.7366	4.8038	4.7714
(*N*=5366)		*0.0288*	*0.0281*	*0.0304*	*0.0330*	*0.0444*	*0.0965*				
	F%=51.79%	3.1426	2.8351	2.5269	2.2462	2.0039	1.9804	LHL	2.9128	2.7837	2.8460
		0.0277	*0.0271*	*0.0304*	*0.0330*	*0.0451*	*0.0922*		*0.0148*	*0.0146*	*0.0100*
2010	M%=48.57%	3.3319	3.0593	2.8200	2.4390	2.2100	2.0703	AVEFLLI	4.5165	4.5727	4.5454
(*N*=9826)		*0.0246*	*0.0231*	*0.0224*	*0.0243*	*0.0300*	*0.0379*				
	F%=51.43%	3.2130	2.8744	2.5388	2.2189	1.9516	1.7190	LHL	2.9248	2.7439	2.8317
		0.0241	*0.0228*	*0.0220*	*0.0252*	*0.0305*	*0.0393*		*0.0114*	*0.0114*	*0.0078*
2012	M%=52.03%	3.2899	3.0642	2.7198	2.3969	2.2067	1.9788	AVEFLLI	4.4471	4.5148	4.4796
(*N*=10024)		*0.0250*	*0.0238*	*0.0206*	*0.0235*	*0.0256*	*0.0314*				
	F%=47.97%	3.1972	2.8004	2.5100	2.1754	1.8754	1.8038	LHL	2.8653	2.6877	2.7801
		0.0234	*0.0232*	*0.0220*	*0.0258*	*0.0302*	*0.0356*		*0.0111*	*0.0112*	*0.0082*
2013	M%=51.10%	3.3700	3.1442	2.8989	2.6594	2.2920	2.1401	AVEFLLI	4.4715	4.5069	4.4888
(*N*=9673)		*0.0230*	*0.0228*	*0.0220*	*0.0242*	*0.0281*	*0.0341*				
	F%=48.90%	3.2845	3.0813	2.7046	2.4313	2.1256	1.9722	LHL	2.9999	2.8806	2.9416
		0.0239	*0.0221*	*0.0234*	*0.0255*	*0.0299*	*0.0363*		*0.0108*	*0.0112*	*0.0080*
2015	M%=47.64%	3.2939	3.1028	2.7807	2.5468	2.3439	2.1640	AVEFLLI	4.3861	4.4110	4.3991
(*N*=9639)		*0.0241*	*0.0260*	*0.0240*	*0.0248*	*0.0256*	*0.0318*				
	F%=52.36%	3.2459	2.9823	2.6259	2.3650	2.0908	1.9608	LHL	2.9111	2.7881	2.8467
		0.0235	*0.0254*	*0.0220*	*0.0248*	*0.0262*	*0.0313*		*0.0114*	*0.0111*	*0.0076*
2017	M%=47.92%	3.2133	2.9754	2.7089	2.3455	2.1448	2.0659	AVEFLLI	4.3612	4.4000	4.3814
(*N*=10908)		*0.0241*	*0.0248*	*0.0238*	*0.0237*	*0.0239*	*0.0292*				
	F%=52.08%	3.1936	2.8760	2.5039	2.2340	1.9220	1.8574	LHL	2.7856	2.6815	2.7314
		0.0224	*0.0240*	*0.0218*	*0.0226*	*0.0231*	*0.0300*		*0.0111*	*0.0105*	*0.0073*

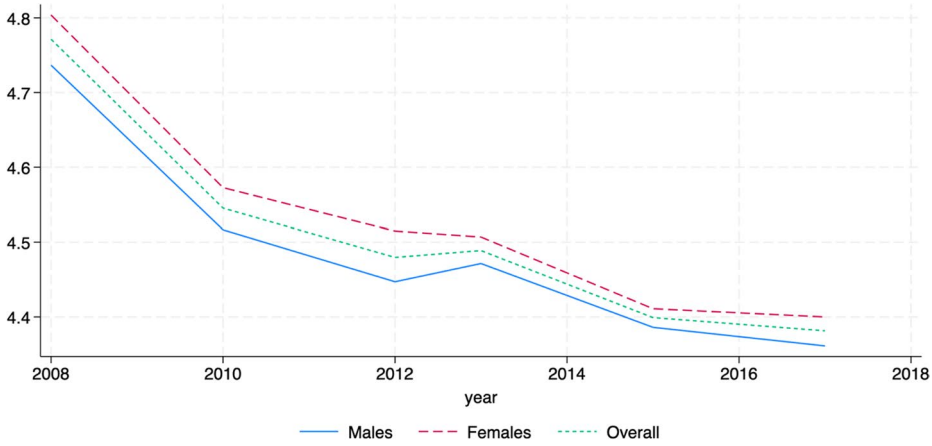

Diagram 2 China Average Expected Future Life Length Index (AVEFLLI), 2008–2017

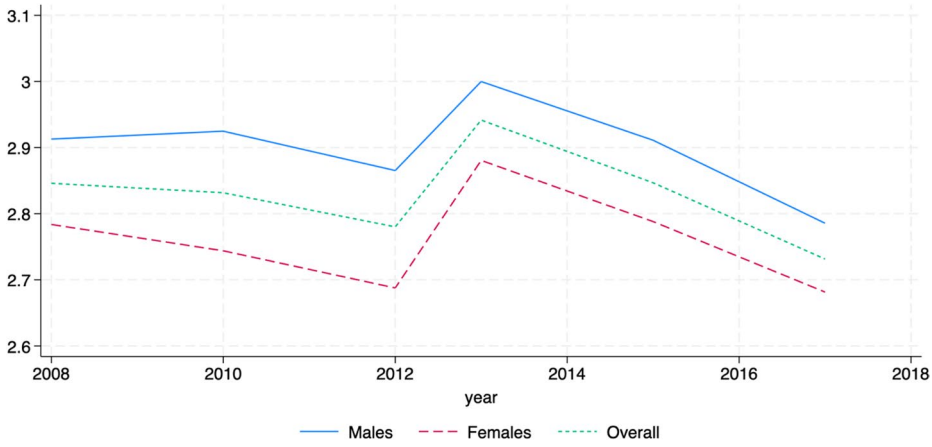

Diagram 3 China Long and Healthy Life Indices (LHL), 2008–2017

That the significant overall decline over the observation period is unambiguous can be seen from the following Table 2[12] which reports that the statistically significant decline was unambiguous over the period in the joint distribution and in the marginal future life length and health distributions. The Panel B result is notable in that, when the analysis is in terms of anticipated future life length, rather than life expectancy at birth and without reference to health, the measures record declines rather than increases for both males and females, the result of the composition effect induced by an increasing proportion of older individuals and a diminishing proportion of younger individuals in the population brought about by population ageing.

[12] These results were based upon Tables 18, 19, 20, 21 reported in the appendix.

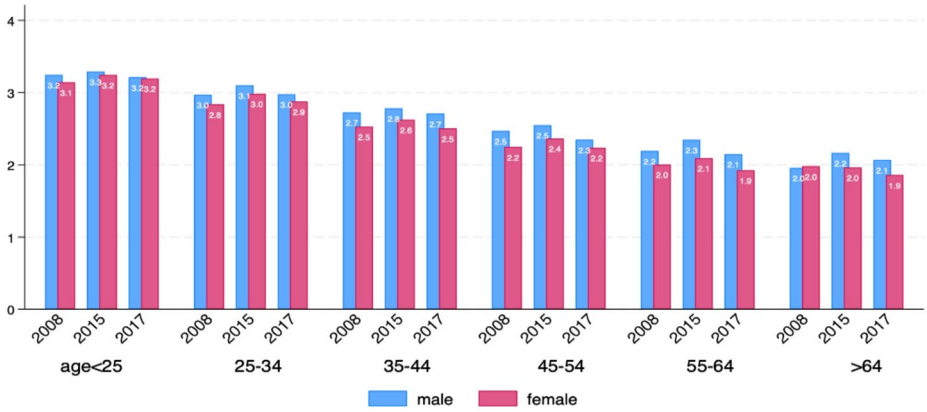

Diagram 4 China Ordered Categorical Mean (OCM) Health Index by age and gender, 2008, 2015 and 2017

Table 2 China First Order Dominance Comparison Joint Distribution, Life Length, and Health

Panel A. First Order Dominance Comparison Joint Distribution			
	$\sum_{k=1}^{K}\sum_{j=1}^{J}\left(F_{i,j}^{2017}-F_{i,j}^{2008}\right)$	$\sum_{k=1}^{K}\sum_{j=1}^{J}\left\lvert F_{i,j}^{2017}-F_{i,j}^{2008}\right\rvert$	AMB=$1-\frac{\sum_{k=1}^{K}\sum_{j=1}^{J}\left(F_{i,j}^{2017}-F_{i,j}^{2008}\right)}{\sum_{k=1}^{K}\sum_{j=1}^{J}\left\lvert F_{i,j}^{2017}-F_{i,j}^{2008}\right\rvert}$
Males	2.4247 (0.0449)	2.4247	0.0
Females	2.7199 (0.0434)	2.7199	0.0
Panel B. First Order Dominance Comparison Life Length			
	$\sum_{j=1}^{J}\left(F_{j}^{2017}-F_{j}^{2008}\right)$	$\sum_{j=1}^{J}\left\lvert F_{j}^{2017}-F_{j}^{2008}\right\rvert$	AMB=$1-\frac{\sum_{j=1}^{J}\left(F_{j}^{2017}-F_{j}^{2008}\right)}{\sum_{j=1}^{J}\left\lvert F_{j}^{2017}-F_{j}^{2008}\right\rvert}$
Males	0.6887 (0.0217)	0.6887	0.0
Females	0.7501 (0.0206)	0.7501	0.0
Panel C. First Order Dominance Comparison Health			
	$\sum_{k=1}^{K}\left(F_{k}^{2017}-F_{k}^{2008}\right)$	$\sum_{k=1}^{K}\left\lvert F_{k}^{2017}-F_{k}^{2008}\right\rvert$	AMB=$1-\frac{\sum_{k=1}^{K}\left(F_{k}^{2017}-F_{k}^{2008}\right)}{\sum_{k=1}^{K}\left\lvert F_{k}^{2017}-F_{k}^{2008}\right\rvert}$
Males	0.2047 (0.0181)	0.2047	0.0
Females	0.2091 (0.0177)	0.2091	0.0

3.2 The Case of the United Kingdom

The Understanding Society[13] dataset, a large-scale longitudinal study was employed in the analysis for the United Kingdom. After incomplete records were excluded a 2-period sample of 75,487 subjects remained with recorded levels of self-reported health, age group and gender categories. The seven health status categories were 1 Completely dissatisfied, 2 Mostly dissatisfied, 3 Somewhat dissatisfied, 4 Neither dissatisfied nor satisfied, 5 Some-

[13] See data appendix.

Table 3 UK survival functions and longevity measures

Year	Gender	Survival Function v_k by Age Group						Longevity (Std Err)
		< 26	26–35	36–45	46–55	56–65	> 65	
2010	Overall	0.8522	0.6967	0.5036	0.3324	0.1794	0.0000	2.5643
	Females	0.8535	0.6926	0.4965	0.3232	0.1727	0.0000	2.5385 (0.0190)
	Males	0.8505	0.7020	0.5126	0.3441	0.1878	0.0000	2.5970 (0.0215)
2018	Overall	0.8705	0.7455	0.5845	0.3970	0.2271	0.0000	2.8246
	Females	0.8734	0.7408	0.5758	0.3891	0.2199	0.0000	2.7989 (0.0225)
	Males	0.8669	0.7513	0.5954	0.4067	0.2359	0.0000	2.8561 (0.0250)

Table 4 UK longevity weighting functions

Year	Gender	Longevity Weighting Function w_k by Age Group					
		< 26	26–35	36–45	46–55	56–65	>65
2010	Females	0.2826	0.2412	0.1957	0.1403	0.0913	0.0488
	Males	0.2780	0.2364	0.1952	0.1425	0.0957	0.0522
2018	Females	0.2632	0.2299	0.1950	0.1516	0.1024	0.0579
	Males	0.2593	0.2248	0.1948	0.1544	0.1055	0.0612

what satisfied, 6 Mostly satisfied, 7 Completely Satisfied. The six age group categories, similar to the Chinese case, were determined as < 26, 26–35, 36–45, 46–55, 56–65 and > 65.

Table 3 considers longevity comparisons and indicates that, somewhat contrary to the World Bank statistics[14], while males have significantly greater longevity than females in 2010 ('t' statistic 2.0389) this is no longer the Case in 2018 ('t' statistic 1.7007). However, longevity has increased significantly for both males and females over the period 2010 to 2018 (respective 't' stats 7.8578 and 8.8424) and results in the corresponding longevity weighting functions reported in Table 4.

Table 5 reports the Health Status Survival Functions of the gender-based age group categories for 2010 and 2018 that are employed in the analysis reported in Table 6 which indicates deterioration of health outcomes over the life cycle for both genders. Though the group-by-group differences are seldom significant, when comparing the health outcome of the youngest versus oldest comparison, results in Panel B yields clear and significant evidence of the health decline.

Combining the health indices of Table 6 with the on the longevity weighting function of Table 4 yields the LHL reported in Table 7. Though neither time differences nor the gender differences are statistically significant, it is clear that the prevalence of Long and Healthy lives is diminishing over time and that males have better Long and Healthy life outcomes than females[15].

3.3 The Case of the United States of America

The General Social Survey (GSS), a series of nationally representative cross-sectional interviews conducted in the United States collected by NORC at the University of Chicago and funded by the National Science Foundation is employed in the United States example

[14] The 2010 and 2018 World Bank Life Expectancy Statistics for the UK were 80 and 81 for males and 82 and 83 for females signalling an improvement in the UNDP's life expectancy based statistic in both cases with females doing better than males.

[15] Note that the 2010 and 2018 World Bank Life Expectancy Statistics for the UK were 80 and 81 for males and 82 and 83 for females signalling an improvement in the UNDP's life expectancy-based statistic in both cases with females doing better than males.

Table 5 UK health status survival functions by age group and gender category

Health Status	Females						Males					
	< 26	26–35	36–45	46–55	56–65	> 65	< 26	26–35	36–45	46–55	56–65	> 65
2010												
1	0.9741	0.9693	0.9555	0.9411	0.9436	0.9442	0.9773	0.9764	0.9666	0.9498	0.9509	0.9502
2	0.9244	0.9114	0.8827	0.8615	0.8584	0.8600	0.9416	0.9226	0.9084	0.8797	0.8703	0.8692
3	0.8074	0.7865	0.7472	0.7029	0.7103	0.7047	0.8459	0.7906	0.7659	0.7321	0.7212	0.7234
4	0.7135	0.7143	0.6707	0.6257	0.6355	0.6036	0.7602	0.7136	0.6820	0.6515	0.6396	0.6274
5	0.5415	0.5707	0.5234	0.4983	0.5090	0.4617	0.5935	0.5468	0.5250	0.5126	0.5032	0.4875
6	0.1555	0.1408	0.1147	0.1048	0.0978	0.0995	0.1877	0.1383	0.1112	0.0850	0.0796	0.0993
7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2018												
1	0.9746	0.9638	0.9617	0.9570	0.9359	0.9594	0.9733	0.9729	0.9786	0.9631	0.9548	0.9584
2	0.9130	0.9017	0.8830	0.8572	0.8368	0.8723	0.9315	0.9133	0.9143	0.8860	0.8647	0.8926
3	0.8001	0.7791	0.7485	0.7008	0.6770	0.7298	0.8327	0.7827	0.7705	0.7348	0.7253	0.7537
4	0.6952	0.6875	0.6591	0.6070	0.5863	0.6203	0.7407	0.6749	0.6647	0.6300	0.6259	0.6596
5	0.5118	0.5049	0.4836	0.4460	0.4421	0.4698	0.5808	0.4774	0.4884	0.4851	0.4829	0.4985
6	0.1553	0.1137	0.0965	0.0856	0.0774	0.0848	0.1965	0.1156	0.0813	0.0686	0.0701	0.0776
7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

The seven health status categories were 1 Completely dissatisfied, 2 Mostly dissatisfied, 3 Somewhat dissatisfied, 4 Neither dissatisfied nor satisfied, 5 Somewhat satisfied, 6 Mostly satisfied, 7 Completely Satisfied

Table 6 UK age cohort health indices

Panel A. Age Cohort Health indices by Gender												
Year	Females						Males					
	<26	26–35	36–45	46–55	56–65	>65	<26	26–35	36–45	46–55	56–65	>65
2010												
Index	4.1164	4.0929	3.8943	3.7343	3.7547	3.6737	4.3062	4.0883	3.9591	3.8105	3.7646	3.7571
Std Err.	*0.0578*	*0.0552*	*0.0504*	*0.0538*	*0.0578*	*0.0540*	*0.0637*	*0.0647*	*0.0576*	*0.0614*	*0.0638*	*0.0582*
2018												
Index	4.0500	3.9506	3.8323	3.6536	3.5556	3.7365	4.2556	3.9368	3.8978	3.7676	3.7237	3.8404
Std Err.	*0.0730*	*0.0716*	*0.0645*	*0.0608*	*0.0639*	*0.0560*	*0.0781*	*0.0852*	*0.0735*	*0.0670*	*0.0705*	*0.0598*
Panel B. Within Year Youngest vs. Oldest Group Differences												
	2010						2018					
	Female Diff			Male Diff			Female Diff			Male Diff		
Difference	0.4426			0.5491			0.3135			0.4152		
Std Err.	*0.0791*			*0.0863*			*0.0920*			*0.0984*		
P(diff < 0)	*0.0000*			*0.0000*			*0.0003*			*0.0000*		

(Davern, 2023). GSS covers a wide range of questions from health, marriage and family to work conditions. For comparison purpose, 2010 and 2018 waves of the GSS are employed. With interviewees aged 18 or older, the sample sizes are 2,044 and 2,348 respectively. Participants answer the question, 'Would you say your own health, in general, is 1 Excellent, 2 Good, 3 Fair, or 4 Poor?'. As with the US sample, six age group categories were determined as < 26, 26–35, 36–45, 46–55, 56–65 and > 65.

Table 8 indicates that longevity is clearly increasing over the period more so for females than for males, (though the difference is not statistically significant) which results in the longevity weighting function reported in Table 9.

Turning to the health measures, Table 10 reports the health status survival functions for the gender based age groups in the respective observation years upon which the long and healthy life indices are based. Table 11 reports the health indices for the gender and age determined subpopulations, the results are interesting since they are somewhat different from the previous two examples. In 2010, health outcomes for females are revealed to be uniformly worse than health

Table 7 UK long and healthy life index

Panel A. Long and Healthy Life Index (LHL)				
	2010		2018	
	Female	Male	Female	Male
Index	3.9590	4.0358	3.8558	3.9574
Std Error	*0.0550*	*0.0622*	*0.0675*	*0.0755*
Panel B. LHL Comparisons				
	Year differences (2018–2010)		Gender Differences (Males-Females)	
	Females	Males	2010	2018
Difference	0.1032	0.0785	0.0768	0.1016
Standard Error	*0.0871*	*0.0978*	*0.0830*	*0.1013*
P(difference < 0)	*0.1180*	*0.2113*	*0.1775*	*0.1579*

Table 8 USA population aging and survival functions and longevity measures

Year	Gender	Survival Function v_k by Age Group						Longevity (Std Err)	
		< 26	26–35	36–45	46–55	56–65	> 65		
2010	Overall	0.9013	0.7022	0.5235	0.3401	0.1693	0.0000	2.6364	0.0834
	Males	0.8945	0.7232	0.5346	0.3287	0.1488	0.0000	2.6298	0.1238
	Females	0.9069	0.6848	0.5143	0.3496	0.1862	0.0000	2.6418	0.1127
2018	Overall	0.8877	0.6943	0.5105	0.3618	0.2100	0.0000	2.6643	0.0753
	Males	0.8721	0.6997	0.5086	0.3563	0.2040	0.0000	2.6408	0.1129
	Females	0.9001	0.6900	0.5121	0.3662	0.2147	0.0000	2.6831	0.1011

Table 9 Longevity weighting functions

Year	Gender	Longevity Weighting Function w_k by Age Group					
		< 26	26–35	36–45	46–55	56–65	>65
2010	Males	0.2755	0.2464	0.1992	0.1473	0.0906	0.0410
	Females	0.2746	0.2490	0.1880	0.1412	0.0960	0.0511
2018	Males	0.2747	0.2395	0.1922	0.1397	0.0979	0.0560
	Females	0.2715	0.2444	0.1873	0.1390	0.0994	0.0583

Table 10 US health status survival functions

Health Status	Females						Males					
	< 26	26–35	36–45	46–55	56–65	> 65	< 26	26–35	36–45	46–55	56–65	> 65
2010												
1	0.9846	0.9613	0.9580	0.9130	0.9298	0.9154	1.0000	1.0000	0.9725	0.9076	0.9038	0.8953
2	0.7231	0.7484	0.7059	0.6783	0.6579	0.6231	0.7869	0.7879	0.8165	0.7983	0.7115	0.5465
3	0.2000	0.3290	0.2689	0.2870	0.2018	0.1923	0.3115	0.3232	0.3119	0.2101	0.2115	0.1744
4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2018												
1	0.9540	0.9617	0.9742	0.9528	0.9318	0.9251	0.9775	0.9750	0.9474	0.9434	0.9434	0.8873
2	0.8276	0.7541	0.7419	0.7008	0.7197	0.6952	0.7416	0.7167	0.7143	0.7264	0.7075	0.6338
3	0.3103	0.2459	0.2129	0.1969	0.2424	0.1925	0.2809	0.2167	0.2481	0.2453	0.2170	0.1972
4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Health Status Categories were 1 Excellent, 2 Good, 3 Fair, or 4 Poor

Table 11 US age cohort health indices

Health Status	Females						Males					
	<26	26–35	36–45	46–55	56–65	>65	<26	26–35	36–45	46–55	56–65	>65
2010												
Index	1.9077	2.0387	1.9328	1.8783	1.7895	1.7308	2.0984	2.1111	2.1009	1.9160	1.8269	1.6163
Std Err.	*0.2478*	*0.1606*	*0.1832*	*0.1862*	*0.1863*	*0.1738*	*0.2558*	*0.2007*	*0.1913*	*0.1832*	*0.1954*	*0.2117*
2018												
Index	2.0920	1.9617	1.9290	1.8504	1.8939	1.8128	2.0000	1.9083	1.9098	1.9151	1.8679	1.7183
Std Err.	*0.2142*	*0.1478*	*0.1605*	*0.1770*	*0.1738*	*0.1456*	*0.2120*	*0.1824*	*0.1732*	*0.1941*	*0.1938*	*0.1662*

Table 12 US long and healthy life index

Panel A. Long and Healthy Life Index (LHL)				
	2010		2018	
	Female	Male	Female	Male
Index	1.9205	2.0308	1.9601	1.9201
Std Error	*0.0922*	*0.0999*	*0.0810*	*0.0871*
Panel B. LHL Comparison				
	Year differences (2018–2010)		Gender Differences (Males-Females)	
	Females	Males	2010	2018
Difference	0.0396	–0.1107	0.1103	–0.0400
Standard Error	*0.1227*	*0.1325*	*0.1359*	*0.1189*
P(|difference|<0)	*0.3735*	*0.2018*	*0.2086*	*0.3683*

outcomes for males at every age group whereas in 2018 things are very different, with females having better health outcomes than males at every age group except for the 46 to 55 year-olds. Over the 8 years health outcomes improved for females except for the 36 to 55 year-olds whereas for males, with the exception of the over 55's, health outcomes deteriorated over the period.

Taken together these results in the LHL for the US reported in Table 12 where it may be observed that things have improved for females and deteriorated for males, resulting in females having inferior long and healthy life prospects to males in 2010 whereas they had superior long and healthy life prospects to males in 2018[16].

3.4 International Comparisons

On the heroic presumption that overall perceptions of acceptable health levels are similar across nations, it is possible to compare nations[17]. To make such between nation comparisons the indices have to be standardized by $K-1$ where K is the number of Health outcome categories (for China, the UK and the USA this was 5,7 and 4 respectively). Table 13 reports the adjusted long and healthy life indices and their corresponding standard errors while Table 14 reports the details of the 3-way statistical comparisons. Assuming similarities in what is perceived as an acceptable health level across nations, and employing a 10% critical value, China appears to have better Long and Healthy Life experiences than the UK in both years and both Genders, it has better Long and Healthy Life experiences in all but 2018 females than the United States. The United Kingdom and the USA are indistinguishable in terms of their long and healthy life outcomes.

However, it is clear that the Long and Healthy Life declined between 2010 and 2018 (except for US females), which has to do with population aging. Due to the global declining population replication rates and increased life expectancy, many developed and developing nations

[16] Note that the 2010 and 2018 World Bank Life Expectancy Statistics for the USA were 79 and 79 for males and 81 and 81 for females signalling no change in the UNDP's life expectancy based statistic in both cases with females doing better than males in both years.

[17] Cross country comparisons of self-reported health can be influenced by factors such as health care provision and the cultural and social experience of well-being. As emphasized by the Functioning and Capabilities Approach, such comparisons inherently reflect differences in cultural and social contexts. In our case, a key distinction among the three countries lies in their health care systems: the UK offers publicly funded health care, the USA relies primarily on private provision, and China occupies an intermediate position (Yip et al., 2019, Cancarevic et al., 2021).

Table 13 Cross country comparison of adjusted long and healthy life index (LHL)

Country	2010		2018	
	Female (Std Err.)	Male (Std Err.)	Female (Std Err.)	Male (Std Err.)
China	0.6860 (0.0028)	0.7312 (0.0028)	0.6704 (0.0024)	0.6964(0.0028)
United Kingdom	0.6598 (0.0092)	0.6726 (0.0110)	0.6426 (0.0112)	0.6597(0.0126)
United States	0.6402 (0.0307)	0.6769 (0.0333)	0.6534 (0.0270)	0.6400(0.0290)

face an aging population which can threaten the longer-term viability of systems which transfer resources from younger to older groups in the population (UN, 2019). This has been recognized in the context of social security and pension systems with concomitantly increasing tax paying/benefit receiving dependency rates which can be challenging for their longer-term fiscal viability (Razin, Sadka, and Swagel, 2002; Heer et al., 2020). In a similar fashion, when health declines with age, a nation's health care system, whose primary objective is promoting the long and healthy lives of its subjects, faces similar concerns since its suppliers of care (the younger healthier cohorts) are diminishing in size relative to its demanders for care (the older less healthy cohorts). The social choice and welfare challenges transcend mere fiscal concerns since they involve increasingly severe real resource constraints. The question is to what extent do health levels deteriorate by age group and, if they do, how would it affect the measurement of the extent to which a nation's subjects live long and healthy lives for public policy evaluation purposes.

4 Conclusions

Promotion of the living of long and healthy enabling lives is an objective of virtually all societies, here self-reported health status recorded at an individual's life stage is employed in examining the extent to which a nation's inhabitants do so. One of the challenges of using self-reported health outcomes highlighted by Bond and Lang (2019) is that they are ordered categorical in nature and the use of artificial cardinalization to facilitate analysis engenders ambiguity in the results that are achieved. Here some techniques for analyzing ordered categorical data that circumvent those ambiguity concerns are developed and a Long and Healthy Life Index (LHL) is proposed and implemented using such information. The approach is more generally applicable whenever a collection of ordered categorical variates with mutual dependencies are confronted (for example, experience and embodied human capital in a human resources index, age, gender and loneliness in a loneliness and aging index etc.). The results demonstrate how data on ordinal variates can be integrated to generate unambiguous and unequivocal cardinally measurable outcome level indices that reflect dependencies between ordinal variates.

These new tools were applied to an analysis of health outcomes in China, during which time the Chinese authorities made great strides in expanding the provision of health care. Contrary to what is observed when sole reliance is placed upon life expectancy data as in the Human Development Index long and healthy lives component (where long and healthy living appears to be on the increase and females appear to be better off than males), the LHL index indicates a significant diminution in the Long and Healthy Lives lived by Chinese subjects with females having significantly inferior outcomes to males. Health outcomes have been shown to unam-

Table 14 Cross country comparison of adjusted long and healthy life index (LHL)

	China-UK				China-USA				UK-USA			
	2010		2018		2010		2018		2010		2018	
	Females	Males	Females	Males	Females	Males	Females	Males	Females	Males	Females	Males
Difference	0.0262	0.0586	0.0278	0.0367	0.0458	0.0543	0.0170	0.0564	0.0196	–0.0043	–0.0108	0.0197
Std Error	*0.0096*	*0.0114*	*0.0115*	*0.0129*	*0.0308*	*0.0334*	*0.0271*	*0.0291*	*0.0320*	*0.0351*	*0.0292*	*0.0316*
"t" stat	*2.7244*	*5.1626*	*2.4270*	*2.8433*	*1.4857*	*1.6249*	*0.6272*	*1.9358*	*0.6116*	*–0.1226*	*–0.3695*	*0.6230*
P(<0)	*0.0032*	*0.0000*	*0.0076*	*0.0022*	*0.0687*	*0.0521*	*0.2653*	*0.0264*	*0.2704*	*0.5488*	*0.6441*	*0.2666*

biguously deteriorate with age and tend to be worse for females than males in a given age group over the comparison period. However, within age group and gender categories, group health status has been maintained over the period. Non-the-less against this backdrop, the LHL index indicates that the Long and Healthy Lives lived by the Chinese population can be seen to be diminishing, a composition effect due to the proportion of the population with inferior health outcomes i.e. the elderly, increasing in size. In effect, though China's increasing investment in healthcare provision has been enough to maintain the health status of age groups, it has not been enough to outweigh the increasing burden that an aging population with greater health care needs has placed upon its relatively shrinking younger population of health care providers. One takeaway is that sole reliance upon the life expectancy at birth of the inhabitants of a nation as an indication of the extent of Long and Healthy Lives that they lead can be misleading since it presumes that subgroup health outcomes have the same distribution throughout society.

When compared to the United Kingdom and the United States over the decade over a similar period (2010–2018 comparisons were used for the UK and the USA given data availability constraints), China seems to be doing relatively well when group health outcomes are integrated into the measures since while their unadulterated life expectancy measures are worse than both the UK and the USA, their Long and Healthy Life experiences are clearly better. From a public policy perspective, these results present serious challenges both real and fiscal.

The ongoing decline in the Long and Healthy Lives of a nation brought about by the low fertility rate engendered aging population, requires response to the changing configuration of a nation's wants and needs and its ability to service those wants and needs. The financial and real burden of care largely falls upon the middle (25–65) age groups which are shrinking proportionately compared to the age groups in need of such care. There needs to be an increased focus on geriatric care and women's health care and an increased focus on technological support for the middle age groups who are supplying that care.

Appendix 1. Inference

Assume t are independently observed populations with n_t observations respectively and let the true m'th outcome level probability $f_{t,m} = p_{t,m}$ for $m = 1, .., M$ be stacked in the $M \times 1$ vector $\underline{p}_t$ and let $\widehat{\underline{p}}_t$ be the corresponding relative frequency estimates of those probabilities based upon a simple random sample. Then, following Rao (2009),

$$\sqrt{n_t}\left(\widehat{\underline{P}_t} - \underline{P_t}\right) \sim_a N\left(0, V\left(\underline{P_t}\right)\right)$$

where:

$$V(\underline{P_t}) = \begin{pmatrix} P_{t,1} & 0 & . & 0 \\ 0 & P_{t,2} & . & 0 \\ . & . & . & . \\ 0 & 0 & . & P_{t,M} \end{pmatrix} - \begin{pmatrix} P_{t,1}^2 & P_{t,1}P_{t,2} & . & P_{t,1}P_{t,M} \\ P_{t,2}P_{t,1} & P_{t,2}^2 & . & P_{t,2}P_{t,M} \\ . & . & . & . \\ P_{t,M}P_{t,1} & P_{t,M}P_{t,2} & . & P_{t,M}^2 \end{pmatrix}$$

Given the *M x M* dimensioned cumulating matrix , where:

$$D = \begin{pmatrix} 1 & 0 & . & 0 \\ 1 & 1 & . & 0 \\ . & . & . & . \\ 1 & 1 & . & 1 \end{pmatrix}$$

$\underline{F_t}$, the *t*'th period vector of CDF values are such that, given $\underline{I}$ is an dimensioned vector of ones:

$$\underline{F}_t = D\underline{P}t \; and \; \underline{S}_t = \underline{I} - D\underline{P}t$$

Each will have variance $DV(\underline{P_t})D\prime$, so that $\sqrt{n_t}\left(\widehat{\underline{F_t}} - \underline{F_t}\right) \sim_a N\left(0,\ DV(\underline{P_t})D'\right)$ and $\sqrt{n_t}(\widehat{\underline{S}_t} - S_t) \sim_a N(0, DV(\underline{P_t})D')$. Since $\mu_{OC,t} = \underline{I'S_t}$, $\sqrt{n_t}(\widehat{\mu}_{OC,t} - \mu_{OC,t}) \sim_a N\left(0, \underline{I'}DV(\underline{P_t})D'\underline{I}\right)$. Note that when relative means are being compared, $\widehat{\mu}_{OC,t}$ and $\mu_{OC,t}$ should be divided by $M-1$ and the variance should be divided by $(M-2)^2$.

Appendix 2. Data

China

Chinese General Social Survey (CGSS) is a national representative continuous survey project initiated in 2003 by the Hong Kong University of Science and Technology and Renmin University. For this project, we require large, representative sample size and specific questions - such as self-reported health. As a result, we use data from the 2008, 2010, 2012, 2013, 2015, and 2017 waves, covering 26 Chinese provinces: Beijing, Tianjin, Hebei, Shanxi, Inner Mongolia, Liaoning, Jilin, Heilongjiang, Shanghai, Jiangsu, Zhejiang, Anhui, Fujian, Jiangxi, Shandong, Henan, Hubei, Hunan, Guangdong, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Shaanxi, Gansu. The provinces not covered include Hainan, Tibet, Qinghai, Xinjiang, and Gansu.

The United Kingdom

Understanding Society: the UK Household Longitudinal Study (UKHLS) is a survey, collecting data from participants by surveying the members of approximately 40,000 households (at Wave 1). Households recruited at the first round of data collection are visited each year to collect information on changes to their household and individual circumstances. It began in 2009 but carries on from the earlier British Household Panel survey, which ran from 1991 to 2008 (University of Essex, Institute for Social and Economic Research, 2022). To this end, the study collects both objective and subjective indicators, which makes it an ideal for the purpose of the project. The samples are constructed by taking yearly cross-sections of the data by region with all the variables of interest mentioned above to compute the different measures described in the paper.

The United States of America

The General Social Survey (GSS) is a series of nationally representative cross-sectional surveys conducted in the United States collected by NORC at the University of Chicago and funded by the National Science Foundation (Davern, 2023). For comparison with China and the UK, we use data from the 2010-2018 waves.

Appendix 3. Background Tables for China

To get a sense of the gender and aging based health status relationships Table 15, Table 16 and Table 17 report the background story. Table 15 indicates that OCM Health Levels appear to decline over the life cycle for both genders, with Male health outcomes first order dominating Female outcomes in almost all age cohorts, suggesting inferior female health status relative to male health throughout the life cycle. Table 16 reports successive cohort comparisons of the OCMs and confirms that successive cohort's health outcomes being first order dominated by preceding cohorts throughout the life cycle for both genders. However, Table 17 suggests little within cohort deterioration of health levels from 2008 to 2017. This is against a backdrop of increasing demand of health care provision (from the older cohorts) and decreasing supply of health care provision (from younger cohorts) in an aging population.

Table 15 Overall male and female cumulative density comparisons by age cohort

Gender Comparison by Age Cohort		Very Unhealthy	Relatively Unhealthy	Normal	Relatively Healthy	Very Healthy	OCM	AMB
<25	Male CDF	0.0062	0.0296	0.1505	0.5206	1.0000	3.2931	0.1271
	Female CDF	0.0047	0.0254	0.1675	0.5880	1.0000	3.2144	
Difference		–0.0015	–0.0042	0.0170	0.0674			
Standard Error		*0.0033*	*0.0073*	*0.0163*	*0.0222*			
*t statistic**		*–0.4556*	*–0.5743*	*1.0398*	*3.0391*			
25–34	Male CDF	0.0089	0.0526	0.2363	0.6465	1.0000	3.0557	0.0000
	Female CDF	0.0126	0.0857	0.2911	0.7014	1.0000	2.9092	
Difference		0.0037	0.0331	0.0548	0.0549			
Standard Error		*0.0046*	*0.0113*	*0.0197*	*0.0209*			
*t statistic**		*0.8024*	*2.9235*	*2.7863*	*2.6233*			
35–44	Male CDF	0.0212	0.1218	0.3359	0.7411	1.0000	2.7800	0.0000
	Female CDF	0.0296	0.1734	0.4267	0.8010	1.0000	2.5693	
Difference		0.0084	0.0516	0.0908	0.0599			
Standard Error		*0.0070*	*0.0158*	*0.0216*	*0.0187*			
*t statistic**		*1.1942*	*3.2615*	*4.1986*	*3.1960*			

Table 15 (continued)

Gender Comparison by Age Cohort		Very Unhealthy	Relatively Unhealthy	Normal	Relatively Healthy	Very Healthy	OCM	AMB
45–54	Male CDF	0.0393	0.1957	0.4597	0.8327	1.0000	2.4726	0.0000
	Female CDF	0.0431	0.2661	0.5432	0.8699	1.0000	2.2777	
Difference		0.0038	0.0704	0.0835	0.0372			
Standard Error		*0.0089*	*0.0188*	*0.0223*	*0.0159*			
*t statistic**		*0.4275*	*3.7486*	*3.7473*	*2.3411*			
55–64	Male CDF	0.0469	0.2625	0.5616	0.8963	1.0000	2.2327	0.0000
	Female CDF	0.0684	0.3531	0.6545	0.9313	1.0000	1.9927	
Difference		0.0215	0.0906	0.0929	0.0350			
Standard Error		*0.0104*	*0.0205*	*0.0217*	*0.0125*			
*t statistic**		*2.0648*	*4.4103*	*4.2746*	*2.7940*			
>65	Male CDF	0.0683	0.3130	0.6151	0.9239	1.0000	2.0797	0.0000
	Female CDF	0.1003	0.4029	0.6767	0.9464	1.0000	1.8737	
Difference		0.0320	0.0899	0.0616	0.0225			
Standard Error		*0.0124*	*0.0213*	*0.0213*	*0.0110*			
*t statistic**		*2.5797*	*4.2118*	*2.8862*	*2.0452*			

*Stoline – Ury 5% c.v. 2.800 1% c.v. 3.289

The results here indicate that male health cumulative densities unambiguously first order dominant Female's except the youngest cohort, indicating inferior female health status relative to male health status throughout the life cycle. OCM Health Levels declines over the life cycle with male levels remaining higher than female levels at every age group. Over the cycle, Male outcomes decline by 36.87% and Female outcomes decline by 41.71%

Table 16 Overall age cohort comparison (Cohort$_T$ – Cohort$_{T-1}$)

Cohort Comparison	OCM Difference	AMB	$F^{(1)}_{1,g'} - F^{(1)}_{1,g}$ **	$F^{(1)}_{2,g'} - F^{(1)}_{2,g}$	$F^{(1)}_{3,g'} - F^{(1)}_{3,g}$	$F^{(1)}_{4,g'} - F^{(1)}_{4,g}$
(25 to 34) – (< 25)	Males Difference	0.0	0.0027	0.0230	0.0858	0.1259
	standard error		*0.0039*	*0.0089*	*0.0176*	*0.0219*
	*t statistic**		*0.6975*	*2.5950*	*4.8864*	*5.7578*
	Females Difference	0.0	0.0079	0.0603	0.1236	0.1134
	standard error		*0.0041*	*0.0102*	*0.0186*	*0.0213*
	*t statistic**		*1.9094*	*5.9384*	*6.6467*	*5.3357*
(35 to 44) – (25 to 34)	Males Difference	0.0	0.0123	0.0692	0.0996	0.0946
	standard error		*0.0054*	*0.0125*	*0.0201*	*0.0205*
	*t statistic**		*2.2619*	*5.5263*	*4.958*	*4.6138*
	Females Difference	0.0	0.0170	0.0877	0.1356	0.0996
	standard error		*0.0064*	*0.0149*	*0.0212*	*0.0192*
	*t statistic**		*2.6496*	*5.8902*	*6.3852*	*5.1861*
(45 to 54)–(35 to 44)	Males Difference	0.0	0.0181	0.0739	0.1238	0.0916
	standard error		*0.0076*	*0.0163*	*0.0217*	*0.0182*
	*t statistic**		*2.3663*	*4.5451*	*5.7017*	*5.0334*
	Females Difference	0.0	0.0135	0.0927	0.1165	0.0689
	standard error		*0.0084*	*0.0184*	*0.0222*	*0.0165*
	*t statistic**		*1.6139*	*5.0376*	*5.2482*	*4.1733*
(55 to 64)–(45 to 54)	Males Difference	0.0	0.0076	0.0668	0.1019	0.0636
	standard error		*0.0091*	*0.0187*	*0.0222*	*0.0152*
	*t statistic**		*0.8370*	*3.5655*	*4.5820*	*4.1732*
	Females Difference	0.0	0.0253	0.0870	0.1113	0.0614
	standard error		*0.0102*	*0.0206*	*0.0218*	*0.0133*
	*t statistic**		*2.4695*	*4.2265*	*5.1108*	*4.6131*
(65 plus)–(55 to 64)	Males Difference	0.0	0.0214	0.0505	0.0535	0.0276
	standard error		*0.0104*	*0.0202*	*0.0220*	*0.0128*
	*t statistic**		*2.0560*	*2.4982*	*2.4344*	*2.1601*
	Females Difference	0.0	0.0319	0.0498	0.0222	0.0151
	standard error		*0.0124*	*0.0217*	*0.0211*	*0.0107*
	*t statistic**		*2.5709*	*2.2996*	*1.0525*	*1.4099*

*Stoline – Ury 5% c.v. 2.800 1% c.v. 3.289

** $F^{(1)}_{m,g'}$ is the health outcome Cumulative Density Function of health category m cohort g.

This table reports successive cohort CDF comparisons of the OCMs. It shows strong evidence of the deterioration of health status over the life span with successive cohort outcomes being First Order Dominated by preceding cohorts throughout the life cycle for both Males and Females. In each instance there is no ambiguity in any comparison

Table 17 Within age cohort OCM health difference over the decade

Year Comparison	Age Cohort					
	< 25	25–34	35–44	45–54	55–64	> 64
Difference Males 2017–2008	–0.0328	0.0060	–0.0173	–0.1252	–0.0463	0.1114
Standard Errors	*0.0376*	*0.0375*	*0.0386*	*0.0406*	*0.0504*	*0.1008*
Difference Females 2017–2008	0.0510	0.0409	–0.0230	–0.0122	–0.0819	–0.1230
Standard Errors	*0.0356*	*0.0362*	*0.0374*	*0.0400*	*0.0507*	*0.0970*

This table compares OCM between 2017 and 2008 for each age cohort. It shows no significant within cohort deterioration of health levels over the decades

Table 18 Joint probability densities

	Health Designation	Age Group (from shortest expected future life span to longest)					
		> 64	55–64	45–54	35–44	25–34	< 25
Males 2008	Very Unhealthy	0.0027	0.0050	0.0062	0.0054	0.0019	0.0015
	Relatively Unhealthy	0.0066	0.0255	0.0286	0.0209	0.0124	0.0031
	Normal	0.0070	0.0383	0.0599	0.0595	0.0549	0.0290
	Relatively Healthy	0.0077	0.0317	0.0723	0.0862	0.0924	0.0746
	Very Healthy	0.0015	0.0128	0.0309	0.0553	0.0781	0.0881
Females 2008	Very Unhealthy	0.0000	0.0047	0.0083	0.0079	0.0047	0.0022
	Relatively Unhealthy	0.0058	0.0248	0.0410	0.0317	0.0223	0.0054
	Normal	0.0076	0.0335	0.0565	0.0648	0.0626	0.0335
	Relatively Healthy	0.0047	0.0259	0.0641	0.0788	0.0968	0.0856
	Very Healthy	0.0004	0.0043	0.0202	0.0443	0.0777	0.0802
Males 2017	Very Unhealthy	0.0086	0.0121	0.0113	0.0044	0.0015	0.0006
	Relatively Unhealthy	0.0329	0.0442	0.0346	0.0178	0.0069	0.0040
	Normal	0.0400	0.0605	0.0584	0.0448	0.0316	0.0205
	Relatively Healthy	0.0396	0.0610	0.0714	0.0737	0.0616	0.0555
	Very Healthy	0.0096	0.0178	0.0281	0.0407	0.0463	0.0603
Females 2017	Very Unhealthy	0.0116	0.0155	0.0093	0.0056	0.0023	0.0005
	Relatively Unhealthy	0.0338	0.0537	0.0470	0.0292	0.0076	0.0018
	Normal	0.0338	0.0611	0.0581	0.0600	0.0368	0.0201
	Relatively Healthy	0.0305	0.0488	0.0655	0.0771	0.0621	0.0632
	Very Healthy	0.0051	0.0106	0.0239	0.0334	0.0403	0.0518

Table 19 Joint cumulative densities

	Health Designation	Age Group (from shortest expected future life span to longest)					
		> 64	55–64	45–54	35–44	25–34	< 25
Males 2008	Very Unhealthy	0.0027	0.0077	0.0139	0.0193	0.0213	0.0228
	Relatively Unhealthy	0.0093	0.0398	0.0746	0.1009	0.1152	0.1198
	Normal	0.0162	0.0850	0.1797	0.2656	0.3348	0.3684
	Relatively Healthy	0.0240	0.1245	0.2915	0.4635	0.6250	0.7333
	Very Healthy	0.0255	0.1388	0.3367	0.5640	0.8036	1.0000
Females 2008	Very Unhealthy	0.0000	0.0047	0.0130	0.0209	0.0255	0.0277
	Relatively Unhealthy	0.0058	0.0353	0.0846	0.1241	0.1511	0.1587
	Normal	0.0133	0.0763	0.1821	0.2864	0.3760	0.4171
	Relatively Healthy	0.0180	0.1069	0.2767	0.4599	0.6463	0.7729
	Very Healthy	0.0184	0.1116	0.3015	0.5290	0.7931	1.0000
Males 2017	Very Unhealthy	0.0086	0.0207	0.0319	0.0363	0.0379	0.0385
	Relatively Unhealthy	0.0415	0.0978	0.1437	0.1659	0.1743	0.1789
	Normal	0.0815	0.1982	0.3025	0.3694	0.4094	0.4345
	Relatively Healthy	0.1211	0.2988	0.4745	0.6151	0.7167	0.7972
	Very Healthy	0.1307	0.3262	0.5299	0.7113	0.8592	1.0000
Females 2017	Very Unhealthy	0.0116	0.0271	0.0364	0.0421	0.0444	0.0449
	Relatively Unhealthy	0.0454	0.1146	0.1709	0.2058	0.2156	0.2179
	Normal	0.0792	0.2095	0.3239	0.4188	0.4654	0.4878
	Relatively Healthy	0.1097	0.2887	0.4686	0.6406	0.7493	0.8349
	Very Healthy	0.1148	0.3043	0.5082	0.7136	0.8627	1.0000

Table 20 Marginal analysis future life length

		Age Group (from shortest expected future life span to longest)					
		> 64	55–64	45–54	35–44	25–34	< 25
Males 2008	PDF	0.0255	0.1133	0.1979	0.2273	0.2397	0.1964
	CDF	0.0255	0.1388	0.3367	0.5640	0.8036	1.0000
Males 2017	PDF	0.1307	0.1955	0.2037	0.1814	0.1479	0.1408
	CDF	0.1307	0.3262	0.5299	0.7113	0.8592	1.0000
Females 2008	PDF	0.0184	0.0932	0.1900	0.2274	0.2641	0.2069
	CDF	0.0184	0.1116	0.3015	0.5290	0.7931	1.0000
Females 2017	PDF	0.1148	0.1896	0.2038	0.2054	0.1491	0.1373
	CDF	0.1148	0.3043	0.5082	0.7136	0.8627	1.0000

Table 21 Marginal analysis health status

		Very Unhealthy	Relatively Unhealthy	Normal	Relatively Healthy	Very Healthy
Males 2008	PDF	0.0228	0.0970	0.2486	0.3649	0.2667
	CDF	0.0228	0.1198	0.3684	0.7333	1.0000
Males 2017	PDF	0.0385	0.1404	0.2556	0.3627	0.2028
	CDF	0.0385	0.1789	0.4345	0.7972	1.0000
Females 2008	PDF	0.0277	0.1310	0.2584	0.3559	0.2271
	CDF	0.0277	0.1587	0.4171	0.7729	1.0000
Females 2017	PDF	0.0449	0.1730	0.2698	0.3471	0.1651
	CDF	0.0449	0.2179	0.4878	0.8349	1.0000

Data Availability The Chinese General Social Survey (CGSS) data that support the findings of this study are publicly available through the Chinese National Survey Data Archive at http://www.cnsda.org/. The *Unders tanding Society* survey data are publicly available at https://www.understandingsociety.ac.uk/. The General Social Survey (GSS) data are publicly available at https://gss.norc.org/.

Declarations

Conflict of Interest Statement The author declares that there are no conflicts of interest regarding the publication of this article.

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